CS619 Prototype Guide

Smart Household and Home Services Management System

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Prototype Phase — Complete Guide

Ye guide aap ke **Proposal**, **SRS**, aur **Design Document** files parh kar banayi gayi hai. Maqsad ye hai ke ab **Prototype** phase ko step-by-step, AI coding tools (Antigravity, Cursor, Claude Code) ke sath, sahi tareeqay se complete kiya ja sakay.

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1. Prototype Actually Hai Kya?

VU CS619 me **Prototype** ka matlab pura final project nahi hota. Ye ek **chalta phirta chhota version** hota hai jo sirf ye prove karta hai ke:

- Aap ka **frontend aur backend connect** ho rahe hain
- Aap ka **database design kaam kar raha hai**
- Aap ne jo **tools/technologies proposal me likhi thin, wo istemal ki hain**
- Basic **CRUD (Create, Read, Update, Delete)** operations chal rahe hain

Aap ki apni file (“Prototype Requirements”) ke mutabiq, is phase me sirf **5 modules** banane hain — poora system nahi. Neeche di gayi list hi aap ka **official scope** hai:

#	Module	Frontend	Backend
1	User Registration & Login	Registration/Login/Profile forms + validation	Users table (customer/provider/admin roles), Auth APIs
2	Household Task Management (CRUD)	Add/Edit/Delete task forms, task list/dashboard	Task model linked to user, CRUD APIs
3	Basic Home Service Booking	Service list, booking form, booking status	Services + Bookings models, booking APIs
4	Basic Admin Monitoring	Simple admin dashboard (users + bookings list)	Admin APIs, basic role-based access
5	Basic System Integration	Navigation between login/dashboard/tasks/booking	Frontend-backend connectivity, basic activity logs

> ***Yaad rahe:*** *Payments, Invoices, Complaints, Feedback, Notifications — ye sab aap ki SRS/DD me hain, lekin* ***prototype scope file me nahi hain****. Prototype me* ***inko chhod sakte hain*** *ya sirf UI ka ek dummy button rakh sakte hain (functional banana zaroori nahi). Agar extra time ho to bana sakte hain, lekin pehle 5 modules zaroori hain.*

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2. Technology Stack (Proposal Se Fixed)

Aap ki proposal me ye tools clearly likhi hain, isliye inhi ko use karna hai:

- **Frontend:** HTML, CSS, JavaScript, Bootstrap, AJAX, jQuery

- **Backend:** PHP framework — **Laravel** (plain/simple PHP allowed nahi hai — proposal me clearly likha hai)
- **Database:** MySQL

AI tool ko yahi stack batana hai, warna wo React/Node ya kuch aur bana dega jo aap ki proposal se match nahi karega.

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3. Prototype Me Kitna “Kaam” Banta Hai?

Chhota lekin **kaam karta hua** system:

- 3 user roles ka basic login (Customer, Provider, Admin) — ya kam se kam Customer + Admin
- Task CRUD fully working (add/edit/delete/view)
- Ek simple booking form jo database me save ho
- Admin dashboard jahan users aur bookings list ho jayen
- Sab pages ek dusre se navigate ho sakein (menu/navbar)

Design/UI perfect hona zaroori nahi — Bootstrap ka default look bhi chalega. Marks **functionality** par milte hain, sirf visuals par nahi.

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4. Recommended Folder Structure (Laravel)

```
project-root/
├─ app/
│   ├─ Http/Controllers/
│   │    ├─ AuthController.php
│   │    ├─ TaskController.php
│   │    ├─ BookingController.php
│   │    └─ AdminController.php
│   └─ Models/
│        ├─ User.php
│        ├─ HouseholdTask.php
│        ├─ ServiceCategory.php
│        └─ Booking.php
├─ database/migrations/
├─ resources/views/
│   ├─ auth/ (login.blade.php, register.blade.php)
│   ├─ tasks/
│   ├─ bookings/
│   └─ admin/
├─ routes/web.php
└─ public/css, public/js
```

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5. Step-by-Step: AI Tool Se Prototype Banane Ka Tareeqa

Ye steps **Antigravity, Cursor, ya Claude Code** — teeno me almost same tareeqay se follow honge, chahe interface thora different ho.

Step 1 — Naya Project Folder Banayen

- Apne computer par ek naya empty folder banayen, misal: `smart-household-prototype`
- AI tool (Antigravity/Cursor/Claude Code) usi folder ko open karein

Step 2 — AI Tool Ko Context Dein

Sabse pehle apni **SRS aur Design Document** files (ya kam az kam unka database table wala hissa) us tool ke context/chat me daalein ya paste karein, taake AI ko field names, roles, aur relationships pata hon.

Step 3 — "Master Prompt" Use Karein

Neeche di gayi prompt seedhi copy-paste kar ke AI tool ko dein (is guide ke saath ek alag `.txt` prompt file bhi di ja rahi hai):

> *"Build a Laravel + MySQL + Bootstrap prototype for a 'Smart Household and Home Services Management System'. Scope: (1) User registration/login with roles customer, provider, admin, (2) Household Task CRUD linked to logged-in user, (3) Home service listing + booking form storing bookings in DB, (4) Simple admin dashboard listing users and bookings, (5) Shared navigation bar linking all pages. Use Laravel's built-in authentication (Breeze), Eloquent models with migrations for Users, HouseholdTasks, ServiceCategories, and Bookings tables matching these fields: [yahan apni DD ki table list paste karein]. Keep the UI simple with Bootstrap. Do not build payments, invoices, complaints, or notifications — those are out of scope for this prototype. Set up the project step by step: first migrations, then models, then controllers, then routes, then blade views, then test each module."*

Step 4 — Ek Ek Module Bana Kar Test Karein

AI se poora system ek sath mat banwayen. Is tarteeb se chalen:

1. Laravel project setup + database connection (.env) + `php artisan migrate`
2. Authentication (Breeze/Laravel auth) — register/login/roles
3. Household Task CRUD — test kar ke dekhein add/edit/delete chal raha hai
4. Service listing + Booking form — DB me save ho raha hai check karein
5. Admin dashboard — users/bookings list dikh rahi hai
6. Navigation bar + final integration testing

Har step ke baad AI se kahen: **"run and verify this works before moving to the next module"** — is se bugs jama nahi hotay.

Step 5 — Local Server Chalayen Aur Check Karein

```
php artisan serve
```

Browser me `http://127.0.0.1:8000` khol kar har form/page manually test karein (jo Test Cases aap ki DD file me diye hain, wahi use karein — TC-001 se TC-005 tak in-scope hain).

Step 6 — Database Verify Karein

phpMyAdmin ya `php artisan tinker` se check karein ke data actually tables me save ho raha hai.

Step 7 — Video/Screenshots Le Lein

VU ko usually prototype ka **working demo video ya screenshots** submit karna hota hai — is liye har module ka short screen recording pehle hi bana lein.

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6. Common Mistakes Se Bachein

- ✕ Plain PHP (bina framework) use na karein — proposal me mana hai
- ✕ Poora SRS ka har feature banane ki koshish na karein (payments, notifications, etc.) — sirf 5 scoped modules
- ✕ AI se ek hi bar me "poora system bana do" na kahen — module-by-module karayen
- ✕ Design/CSS par zyada waqt zaya na karein — functionality priority hai
- ✓ Database schema wahi rakhein jo Design Document me diya gaya hai (Customer, ServiceProvider, Admin, ServiceCategory, HouseholdTask, Booking…)

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7. Deliverable Checklist (Submission Se Pehle)

- [ ] Laravel project working (login/register/tasks/booking/admin)
- [ ] MySQL database with correct tables
- [ ] Source code zip (bina `node_modules` / `vendor` ke — sirf project code)
- [ ] Short demo video ya screenshots
- [ ] README/short note: “how to run” (composer install, migrate, serve)

Ye guide aap ki upload ki gayi Proposal, SRS, aur Design Document files se derive ki gayi hai. Agar VU ka official Prototype template/marking-criteria document alag se mila ho, to us ke instructions ko priority dein.